End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech (CSE, IT, CSE-AI), Semester: III
(Nov-Dec, 2022)

Subject Code: BAS-201 **Subject: MSE Material Sc and Engg**
Time :3 Hours **Maximum Marks :60**
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 **(5*4=20)**

(a) Mention three Differences between metal and polymers along with some examples?
(b) Lead in superconducting state has critical temperature of 6.2 K. The critical magnetic field at 0K is 0.064 μAm^{-1}. Determine the critical field at 4K?
(c) What are the characteristics of an implant material? Give some examples of polymer implant materials along with their area of application?
(d) The lattice parameter for Fe is 0.2866 nm. What is the minimum angle of first order diffraction if a monochromatic radiation of 0.1790 nm is used?

UNIT-I

Q2 (a) What are composite materials? Explain different composite matrix and their uses in different fields and appliances? **(5,5)**
(b) What are ceramic materials? How is it different from other materials? Tabulate its general properties?

Q3 (a) What are polymers? Justify their use in different industries with the help of suitable examples. **(5,5)**
(b) Write a short note on the electrical and thermal conductivity of different materials such as metal, alloys, ceramics, polymers and composites?

UNIT-II

Q4 (a) Two parallel plates having equal and opposite charges are separated by 2 cm thick slab of dielectric constant 3. If the electric field inside the material is 10^6 V/m. Calculate the Polarization vector and electric flux density? **(5,5)**
(b) What are Piezoelectric Materials? Which inherent property of crystals becomes a reason for piezoelectricity (demonstrate using diagram)? Also mention reverse piezoelectric effect. (with examples)

Q5 (a) A square germanium plate of parameters 10 mm, 10 mm, and thickness 1 mm, is placed in a magnetic field of 1.5 Wb/m^2 acting perpendicular to its thickness. If 1 mA current flows along its length, calculate the Hall voltage developed if the Hall coefficient is 3.5×10^{-4} m^3/coulomb. **(5,5)**
(b) What is the reason of magnetization of a material? Classify between the magnetic materials such as dia-, para-, ferro-, and anti-ferro-magnetic material depending upon their spin.

UNIT-III

Q6 (a) What is an LED? How is it constructed? Explain the basic constituents of an LED light? **(5,5)**
(b) What are photonic crystals? Explain the construction and properties of 1D, 2D, and 3D photonic crystals and their applications.

Q7 (a) What are shape memory alloys? Explain the phenomenon and how it works? **(5,5)**
(b) Explain chromic materials and their widely known variants like thermo-, photo-, and electro-chromic materials?

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UNIT-IV

Q8 What is X-ray diffraction technique? Explain its principle including Bragg's law. Also Explain the Construction, and Working of an X-ray diffractometer (From the generation of X-rays to the selection of single wavelength, interaction with sample, and detection) with the help of a diagram (Bragg's geometry). **(10)**

Q9 What is Raman Spectroscopy? Explain its principle with the help of energy level diagram, Rayleigh, stokes and anti-stokes lines, why stokes lines are higher in intensity? Also explain the construction of a Raman setup with the help of a diagram. (7,3) **(10)**

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